

Sustainability Meets Profitability: A Data-Driven Study of an Eco-Friendly Café

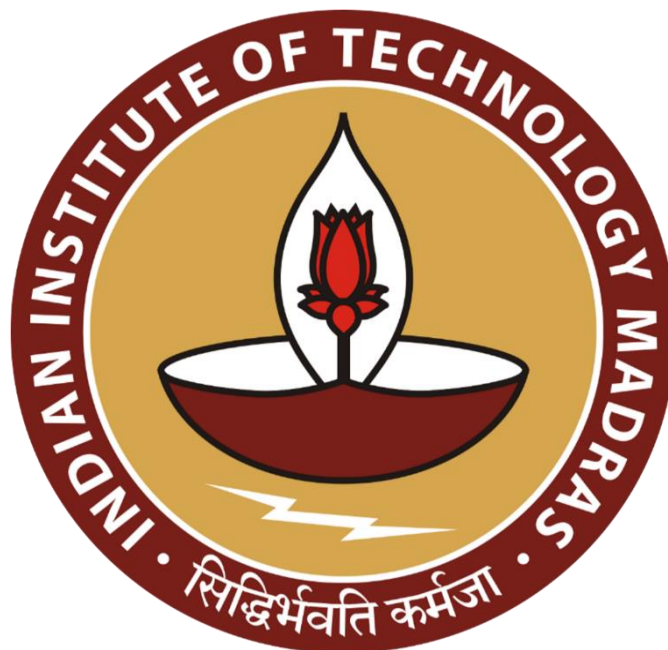
The Final Report for the BDM Capstone Project

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Contents

Topic	Page Number
1. Executive Summary	2
2. Detailed Explanation of Analysis Process	3
3. Results and Findings	9
4. Interpretation of Results and Recommendations	18
5. Summary	21

1. Executive Summary

This project, **Sustainability Meets Profitability: A Data-Driven Study of an Eco-Friendly Café**, focuses on **Sprout**, an eco-friendly café located in Muscat, Sultanate of Oman. It operates as a B2C business in the food and beverage sector, offering healthy and wholesome meals, mostly plant-based, along with fresh juices and smoothies. Sprout follows eco-friendly practices and provides sustainable packaging. The major challenges faced by the café include heavy dependence on an external delivery app, high packaging costs, and lack of customer review analysis. Although the initial report proposed analyzing packaging costs, a significant portion of Sprout's operating expenses due to its commitment to sustainable materials, the café currently lacks relevant data in this area. Consequently, this aspect of the analysis was discarded, and the study redirected its focus towards examining other factors affecting business performance instead.

To analyze these issues, relevant data was collected. After performing various analyses on this data, the key findings were as follows: the average number of orders per day is 50, the average daily sales are OMR 188, and the average sales per order are OMR 3.71. The average commission fee paid per day is OMR 73.24, and the current percentage of the subtotal that Sprout retains is 63%. Customer satisfaction is heavily skewed toward positive feedback. The mode is "Amazing," appearing in 70% of the reviews, indicating that the vast majority of customers are highly satisfied. Other satisfaction levels occur much less frequently: Very Good (8%), Good (7%), and OK (15%). The most common issue reported is Taste/Quality complaints, which account for 41% of all complaints, followed by Wrong/Missing order complaints at 29%. Missing cutlery complaints represent 13%, Less Quantity complaints 10%, and Price complaints 7%. All of this analysis, as well as the more detailed breakdowns, was conducted using Excel.

The data revealed Sprout's near-total reliance on the external delivery platform, Talabat, with direct pickup orders virtually absent and Pro users driving the majority of demand. Daily revenue is highly volatile, and operational peaks are concentrated during lunch and dinner hours, while mid-afternoon and late-evening periods remain consistently low. Minor readying delays of 1-2 minutes dominate, with rare long-tail delays of up to 50 minutes, subtly impacting customer satisfaction. There is low overall cancellation rate. The reviewer base is largely composed of one-time users, reflecting a transactional audience rather than a loyal one.

Financial analysis underscores the structural drain caused by Talabat's commission and marketing fees, limiting net income. Operational interpretation shows inefficiencies during high-pressure periods, particularly in order preparation and delivery timing, while low-demand hours remain underutilized. Customer review analysis points to recurring internal issues such as taste, quality, and packaging errors, rather than platform-related problems. The recommended actions focus on staffing optimization, off-peak promotions, review monitoring, standardized packaging, and renegotiating commissions or incentivizing pickups, targeting improved operational efficiency, enhanced customer experience, and sustainable profitability.

2. Detailed Explanation of Analysis Process

2.1 Data Cleaning and Preprocessing

The data was collected based on the problem statement identified. The data was then cleaned, and some preprocessing was done, before any kind of analysis could be performed on it. Some entries also contained data-type errors, which were corrected to restore consistency across the dataset. Several issues had to be addressed in the raw data, including missing values, errors, duplicates, and inconsistent formats.

The original sales data was split across two separate files, one for September and one for October. Both files were consolidated into a new workbook, ***Sprout_SalesDataAnalysis_Sep2025-Oct2025***, which contains multiple sheets dedicated to different parts of the analysis. The combined dataset was then cleaned and prepared for use.

Key transformations included:

- **RealDate** was extracted from OrderRecievedAt using INT(OrderRecievedAtRange), as the date was given in yyyy-m-dd MM:SS format.
- **OrderHour** was extracted from OrderRecievedAt using MOD(OrderRecievedAtRange,1).
- **AcceptedTime**, **ReadyTime**, **DeliveryTime** and **EstimateReady** were similarly calculated using AcceptedAt, ReadyToPickUpTime, DeliveredAt and EstimatedReadyToPickUpTime respectively.
- **ReadyDelay(mins)** was calculated as (ReadyTime - EstimateReady)*1440, **PrepTime(mins)** was calculated as (ReadyTime - AcceptedTime)*1440 and **DeliveryDelay(mins)** was calculated as (DeliveryTime - EstimateDelivery)*1440
- **Delayed?** Was calculated using IF(ReadyDelay(mins)>0,TRUE)
- **DeliveryDelayed?** Was calculated using IF(DeliveryDelay(mins)>0, TRUE)
- Missing values in **EstimatedReadyToPickUpTime** were filled using the **Average Preparation Time (mins)** and **Date** columns from the *Sprout_preparationTimePerDay* sheet.

A new dataset, ***Daily_SalesSummary***, was generated using a Pivot Table using the cleaned *Sprout_SalesDataAnalysis_Sep2025-Oct2025* data:

The Pivot Group feature was used to group data by day. **Total Orders** per day were calculated by counting OrderID. **Total Sales** per day were computed using the sum of PayoutAmount. **Average Sales per Order** was calculated as Total Sales/Total Orders and **Total Commission Fee Paid** per day was calculated as the sum of commission paid on all orders daily.

The *sprout_reviews* dataset did not contain a complaint category column. To address this:

Reviews with satisfaction levels “OK” and “Good” were filtered. A manual classification column for complaint categories was added. It was observed that reviews marked as “Amazing” and “Very Good” did not contain any complaints.

Through these operations, three fully cleaned and structured datasets were created. These formed the foundation for all subsequent analysis. All cleaning and preprocessing tasks were performed exclusively using Excel.

Data cleaning is a critical step as unclean or inconsistent data can lead to misleading and unreliable insights. With the cleaning process completed, the next stage of the project focuses on analyzing the prepared datasets.

2.2 Detailed Analysis Process

The analytical framework of the report centers on three key dimensions:

- ◆ **Talabat-Related Expense Analysis** – assessing financial dependency on the external delivery platform and evaluating potential savings through alternative options.
- ◆ **Business Performance Analysis** – examining how operational factors affect overall profitability.
- ◆ **Customer Review Analysis** – identifying common pain points and trends in customer satisfaction.

Together, these components form a comprehensive assessment of Sprout’s business model. The analysis employs both quantitative and qualitative methods, with findings visually represented through data visualizations for clarity and actionable insights.

2.2.1 Talabat Expenses Analysis

- **Performance Indicator**

Percentage of Revenue from Talabat	Commission Rate Per Order	Potential savings with Direct Delivery (excluding ads & marketing fee)	Potential savings with Direct Delivery (including ads & marketing fee)	Average Earnings per Order (OMR)	Current percentage of subtotal sprout earns
99%	24.5%	27%	34%	3.71	63%

Table 2.2.1.1: Talabat Dependency

Values for the above table were calculated as follows:

- a. Percentage of Revenue from Talabat: =SUMIFS(Subtotal Range, DeliveryTypeRange, "Platform delivery") / SUM(SubtotalRange))
- b. Commission: =CommissionRange/SubtotalRange (percentage)

- c. Potential savings with direct delivery (excluding ads/marketing fee):

$$=(\text{SUM}(\text{OnlinePaymentFeeRange}) + \text{SUM}(\text{CommissionRange}) + \text{SUM}(\text{VendorChargesRange})) / \text{SUM}(\text{SubtotalRange})$$
- d. Potential savings with direct delivery (Including MarketingFee and Adsfee):

$$=(\text{SUM}(\text{OnlinePaymentFeeRange}) + \text{SUM}(\text{CommissionRange}) + \text{SUM}(\text{VendorChargesRange}) + \text{SUM}(\text{AdsFee}) + (748.52)) / \text{SUM}(\text{SubtotalRange})$$
- e. Average Earnings per order (PayoutAmountRange/TotalOrders):

$$=11441.079/3083$$
- f. Current percentage of subtotal Sprout currently earns

$$=\text{SUM}(\text{EstimatedEarningsRange})/\text{SUM}(\text{SubtotalRange})$$

The overall business analysis reveals a fundamental strategic vulnerability: a 99% reliance on the external platform where the high commission rate erodes margins. The current percentage of subtotal sprout earns is 63%. Potential savings (excluding Ads and Marketing fee) with direct delivery and discontinuing Talabat's delivery, Sprout can earn up to 90% of the subtotal amount and 97% if ads and marketing fee are included. This clearly demonstrates how the high commission fee charged by Talabat is drastically eroding Sprout's net earnings.

- **Platform vs. Pickup Earnings & Share**

Total orders from 1st September to 31st October were 3,088. A pivot table was created with RealDate as rows and DeliveryType as columns, using the sum of PayoutAmount as the values, and grouped by date to calculate earnings for both categories.

Another pivot table was created to determine the share of both delivery types using the count of DeliveryType. The total payout amount for the given period was OMR 11,455.139 (Breakdown: Pickup – OMR 137.555, Platform – OMR 11,317.584). A line chart and a pie chart were generated to visualize these results respectively.

- **Highest Profit Drain Reason**

A profit drain table was manually created with entries including the sum of OnlinePaymentFee, Commission, VendorCharges, AdsFee, MarketingFee, and PlatformFundedVoucher. A column chart was then generated to show the expenses incurred under each category.

- **Pro vs. Not Pro Customers**

From the total 3,088 orders, filters were used to determine the daily count of Pro vs. NoPro customers, which was then used to create a column chart comparing the numbers between both categories. A summary of this data was also used to generate a pie chart illustrating the share contributed by each category.

2.2.2 Business Performance Analysis

- **Peak Delivery Time or Busiest Hours**

Out of the total 3,088 orders, 5 orders were missing OrderReceivedAt data, making it impossible to calculate OrderHour for them. Therefore, these 5 orders were excluded from the analysis, and the remaining 3,083 entries were used. These 5 orders were part of the 23 cancelled orders and could afford to be ignored while doing this particular analysis. A pivot table was then created with RealDate as columns and OrderHour as rows, using the count of OrderID grouped by hour. This was used to generate a column chart comparing the distribution of order-received times across both months. From this, peak active hours and silent hours were identified.

- **Cancellation Reason Share & Cancellation Rate**

Out of the total 3,088 orders, 23 were cancelled, leaving 3,065 orders for analysis. Each category of CancellationReason was summed to obtain the values needed to create a pie chart showing the distribution. For the cancellation rate, a pivot table was prepared to compare the total cancelled orders with the delivered orders across both months. A column chart was then created to illustrate this comparison. The cancellation rate was calculated using the formula Total Cancelled Orders/Total Orders (23/3088). The cancellation rate was found to be only 0.74%.

- **Correlation between Discount and Estimated Earnings**

Correlation was calculated using the formula =CORREL(EstimatedEarnings, TotalDiscount). The resulting value was -0.138721595, approximately -0.13. A scatter plot was created with discount funded by the vendor on the x-axis and net earnings per order on the y-axis. A linear trend line was then added, with both the equation and the R-squared value displayed on the chart. The insights obtained from this analysis are discussed in the Results and Findings section in the next part.

- **Readying Delay**

Out of the total 3065 Orders (3088 -23 cancelled), there were 1363 orders that faced readying delay. Two column charts were created to summarize this. Figure 3.11 shows the total orders that were delayed along with the time they were delayed by in minutes. Figure 3.12 shows the spread of these delays across September and October separately. The charts pinpoint the issue with extreme precision. Figure 3.13, a line chart, is the most insightful metric, demonstrating a clear systemic bias towards delay in the production workflow. Detailed insights are discussed in the next section.

- **Hourly Delays**

From the total 1365 readying delayed orders, a column chart was created, using pivot charts, to depict the number of orders that were facing readying delay, per hour. This hourly delayed visualization is helpful to see what time periods experience the highest readying delay problems.

- **Delivery Delays**

Of the total 3065 orders, 1849 orders were delayed by delivery.

A column chart and a line chart were created to show this. The column chart shows the number of orders that were delayed per hour for both months separately, while the line chart shows the time in minutes that these orders were delayed by.

- **Avoidable Delays**

Total Avoidable Delays	Total Avoidable Cancellations	Average Daily Delay Count	Rate
112 (over 61 days): Breakdown: 61-Bowsher & 51 Mawalih	7 (over 61 days)	1.36 (Orders/Day:153/112)	6.25% of Delayed Orders
Status: Moderate Volume	Status: Excellent	Status: Area for Improvement	Status: Monitor Closely

Table 2.2.2.1: Avoidable Cancellation & Delays Table

The above values were calculated using Excel functions applied to data from the cleaned **Sprout_orderRejectionsPerDay_RawData** and **Sprout_preperationTimePerDay_RawData** files.

- **Daily Sales Summary**

Excel SUM, MAX and MIN were used to get the values below:

Summary	SEPTEMBER 2025	OCTOBER 2025
Sum of Total Orders	1584	1504
Sum of Total Sales	5787.71	5667.42
Sum of Commission Fee Paid	2267.63	2200.06
Sum of Average Sales Per Order	109.41	117.179
Sum of Pro Customers	1156	1074
Sum of Non-Pro Customers	428	430

Table 2.2.2.2: Sum of Daily Sales

Summary	SEPTEMBER 2025	OCTOBER 2025
Max of Total Orders	78	77
Max of Total Sales	299.62	289.76
Max of Commission Fee Paid	117.00	112.09
Max of Average Sales Per Order	4.24	4.55
Max of Pro Customers	64	57
Max of Non-Pro Customers	24	24

Table 2.2.2.3: Max of Daily Sales

Summary	SEPTEMBER 2025	OCTOBER 2025
Min of Total Orders	19	14
Min of Total Sales	72.05	47.04
Min of Commission Fee Paid	27.72	17.98
Min of Average Sales Per Order	2.98	3.20
Min of Pro Customers	12	7
Min of Non-Pro Customers	4	6

Table 2.2.2.4: Min of Daily Sales

2.2.3 Customer Review Analysis

There were total 340 Talabat reviews for the time range 1st April 2024 to 5th October 2025. Out of these 340 reviews, 69 had complaints. This range was taken for Figures 3.16, 3.17 & 3.20. While for figure 3.18 & 3.19, a one year range was considered (November 2024 to October 2025). These constituted 113 reviews out of the 340 total and 44 out of 69 complaints.

Statistic	Satisfaction	Category of Complaint
Unique values	4	5
Mode	Amazing	Taste/Quality complaint
Frequencies	Amazing-70% Very Good-8% Good-7% OK-15%	Less quantity complaint-10% Missing cutlery complaint-13% Price Complaint-7% Taste/Quality complaint-41% Wrong/Missing order complaint-29%

Table 2.2.2.5: Descriptive Summary of Reviews

- **Satisfaction Levels**

A pie chart was created to depict the share of each satisfaction category. The values were obtained by filtering and counting entries from each category. Additionally, a column chart was created to illustrate the varying satisfaction levels of customers throughout the year.

- **Share of Complaint Categories**

Categories were created and manually entered by filtering the OK and Good Satisfaction categories. A pie chart was then generated to show the share of each complaint category, using filtered counts for each category. A column chart was also created to illustrate the distribution of complaints across the year.

- **Customer Feedback Engagement**

The count function was used to determine the total number of customers who left a review once and those who left reviews more than once. The highest number of reviews by a single customer was found to be 13. This information was then visualized using a pie chart.

3. Results and Findings

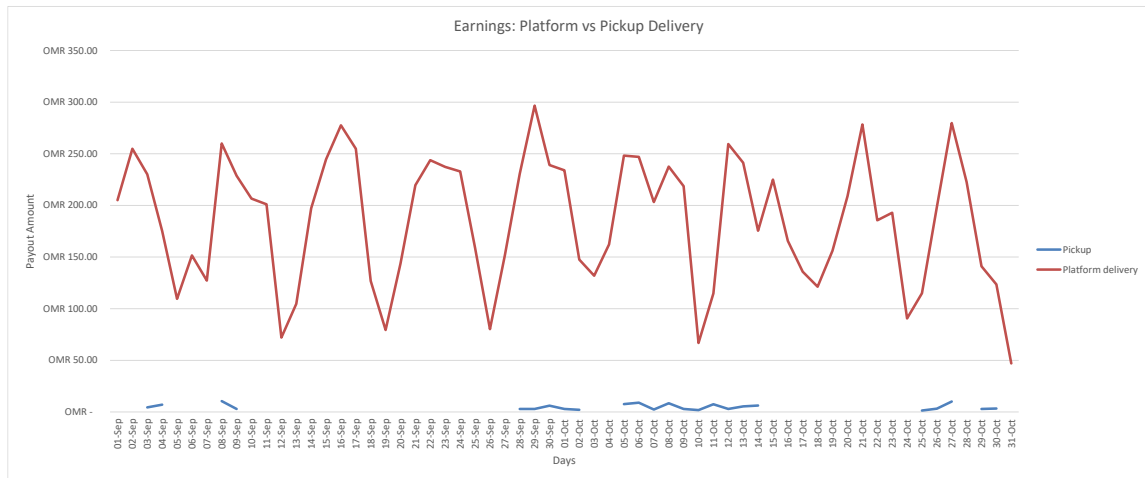


Figure 3.1: Line Chart Showing Earnings by Platform Delivery and Pickup Delivery

The above chart visualizes the daily payout performance over a two-month period (September 1st to October 31st), revealing an overwhelming reliance on third-party logistics compared to direct customer interaction. The Platform delivery series (in red) represents the primary revenue engine, exhibiting high volatility with earnings fluctuating wildly between roughly OMR 50 and nearly OMR 300 per day. This suggests the business is subject to the variable demand cycles and user base of the external app. In stark contrast, the Pickup stream (in blue) is negligible, appearing as a broken, flat line hovering near zero with frequent gaps, indicating that direct consumer takeaway sales are virtually non-existent and that the business model is currently almost entirely sustained by the external delivery partner.

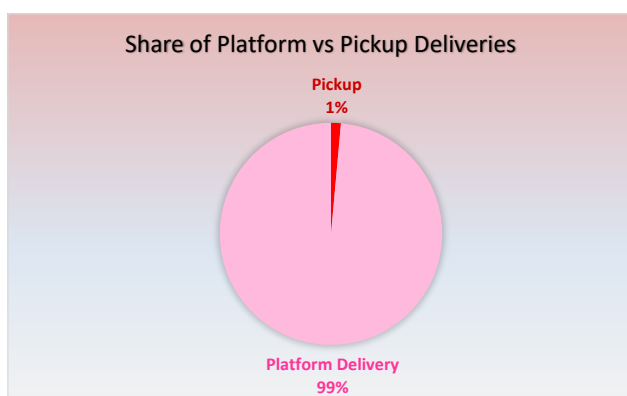


Figure 3.2: Pie Chart Showing Share of Pickup & Platform Delivery

This pie chart serves as a stark visualization of revenue concentration, confirming an almost total dependence on the external service provider. By attributing 99% of the total share to Platform Delivery, the graph illustrates that the business is not operating as a hybrid model but is effectively functioning as a "ghost kitchen" or virtual brand solely reliant on the external platform's user base and logistics. The 1% share for Pickup indicates that direct consumer channels are virtually non-existent, meaning the business has no significant independent footfall or direct customer loyalty. This highlights a critical strategic vulnerability: without a diversified sales mix, the business is entirely exposed to the commission structures, algorithm changes, and operational continuity of the external platform.

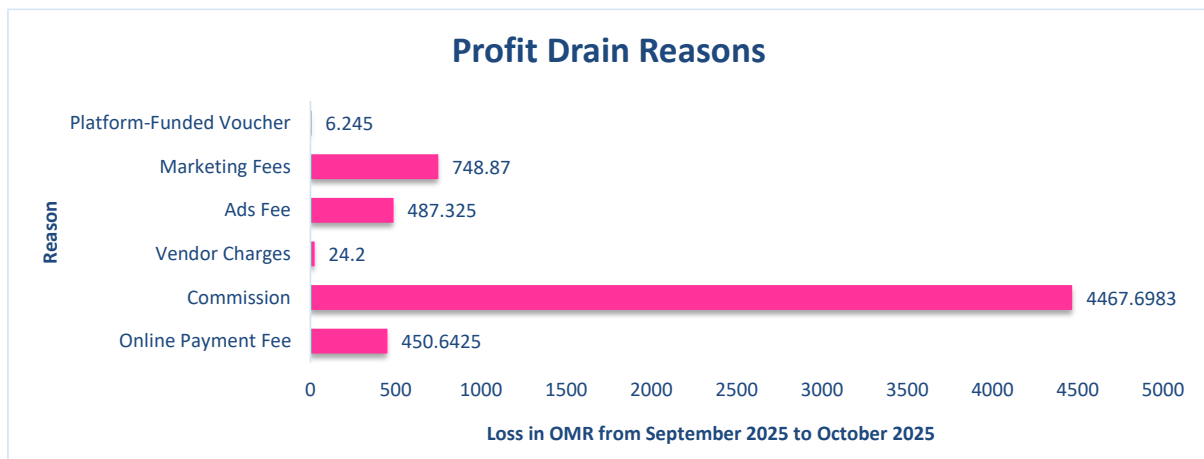


Figure 3.3: Bar Chart Showing Profit Drain from each Category

This bar chart breaks down the operational costs incurred from September to October 2025, identifying Commission as the overwhelmingly dominant factor in profit erosion, at approximately OMR 4,467, the commission fees dwarf all other expense categories combined, acting as a massive tax on the business's revenue. This confirms the high price of relying on the external platform: the business is paying a steep premium for the platform's logistics and customer access. Furthermore, the combined total of Marketing Fees OMR 748, and Ads Fees OMR 487, indicates that simply being on the platform isn't enough, the business is also spending heavily (~OMR 1235) just to remain visible within the app's algorithm. When combined with Online Payment Fees of OMR 450, the chart reveals a cost structure where the external platform captures a disproportionate share of the value, leaving the café with significantly squeezed margins despite high sales volume.

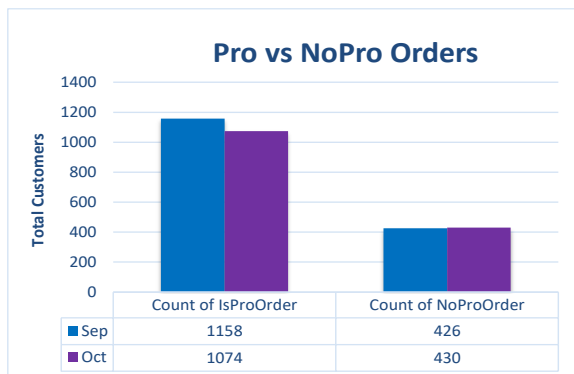


Figure 3.4: Column Chart Showing Count of Pro vs. Not Pro Customers Monthly

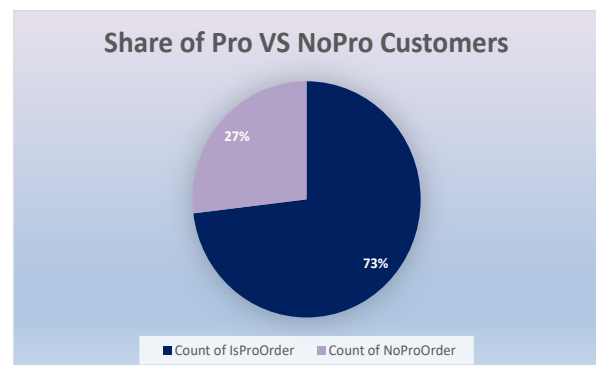


Figure 3.5: Chart Showing Share of Pro vs. Not Pro customers

The above charts reveal that the business is overwhelmingly reliant on customers subscribing to an external platform's Pro service, which comprises 73% of the unique customer pool versus only 27% who are Not Pro users. This Pro segment drives the sales engine, as shown by their 1,158 orders in September compared to just 426 orders from NoPro users. While this high order volume (which repeats with 1,074 orders in October) demonstrates a stable, high-frequency

customer base, the heavy dependency on users whose primary loyalty and incentive (such as free delivery) are tied to the external platform's pricing structure, presents a significant strategic risk, rather than reflecting inherent loyalty to the vendor itself.

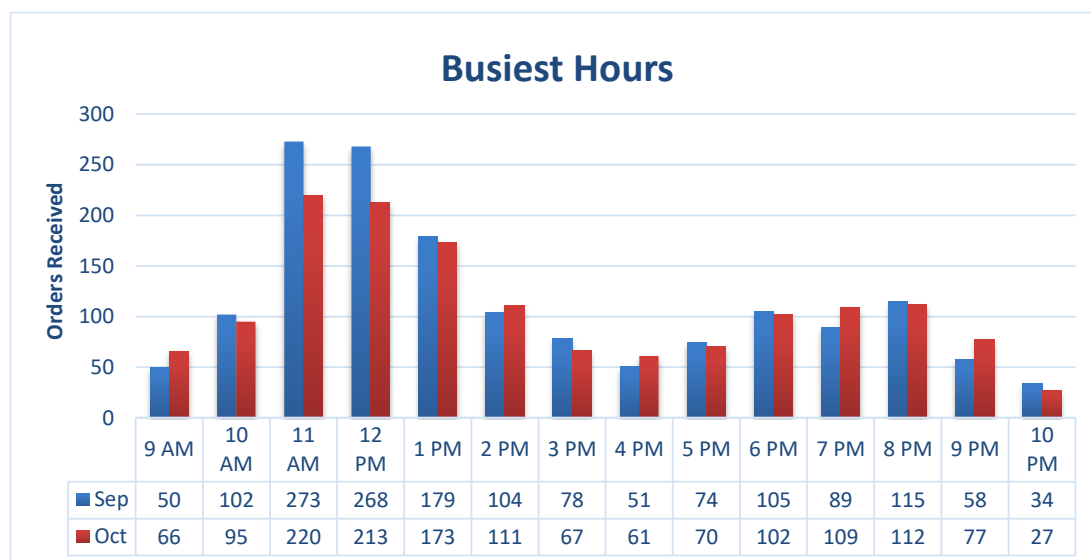


Figure 3.6: Column Chart Showing Hourly Orders Received Across September and October

This column chart dissects the operational rhythm of the business, confirming two distinct peak periods for orders received throughout the day. The dominant peak is the Lunch Rush (11 AM to 12 PM), with September volumes exceeding 260 orders in each hour, drastically overshadowing the secondary Dinner Rush (7 PM to 9 PM) where order volumes hover consistently around 100 to 115 orders. Crucially, the data reveals a significant month-over-month decline in October, particularly during the critical lunch window, (combined drop is ~108 orders), which lost orders compared to September (falling from 273 to 220 at 11 AM and 268 to 213 at 12 PM). This concentrated drop in volume during the busiest operational period is the primary concern, indicating that the overall revenue dip previously observed is entirely attributable to losing high-volume business during the highest-stress time slots, while the mid-afternoon (3 PM-5 PM) remains the consistent period of lowest demand.

Based on the provided chart, the less active hours for receiving orders generally occur during two main periods in both September and October. The quietest period is typically the mid-to-late afternoon, specifically between 3 PM and 5 PM, where order volumes consistently drop to their lowest daily levels, hovering around 50 to 80 orders. Additionally, the period immediately following the evening rush, around 9 PM and 10 PM, also constitutes a period of low activity as order counts drop sharply to their absolute lowest point, often between 30 and 80 orders, marking the close of the ordering day.

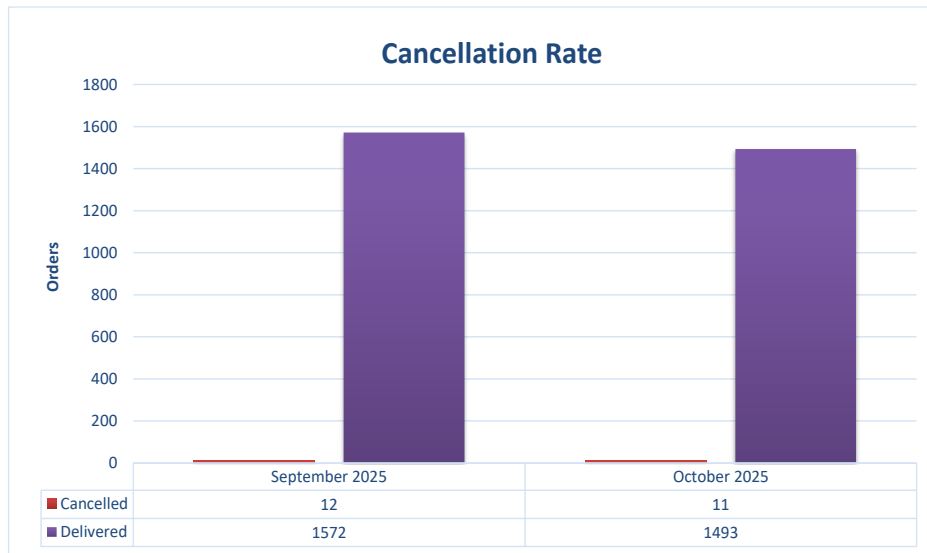


Figure 3.7: Cancellation vs. Delivered Summary of Two Months

This chart presents a highly positive operational metric, showing that the number of hard order cancellations is extremely low compared to the volume of delivered orders. In September, only 12 orders were cancelled out of 1,584 total (a rate of approximately 0.76%), which slightly improved to 11 cancellations out of 1,504 orders in October (a rate of approximately 0.73%). While this -1% cancellation rate is excellent, it must be viewed in the context of the high level quality issues identified below, where logistical failures (delay or unreachable customers) accounted for over half of all cancellation reasons. This drop suggests that the delivery ecosystem or the café is successfully resolving the vast majority of these service issues before they result in a final, recorded cancellation, even as total order volume declined by 80 orders month-over-month.

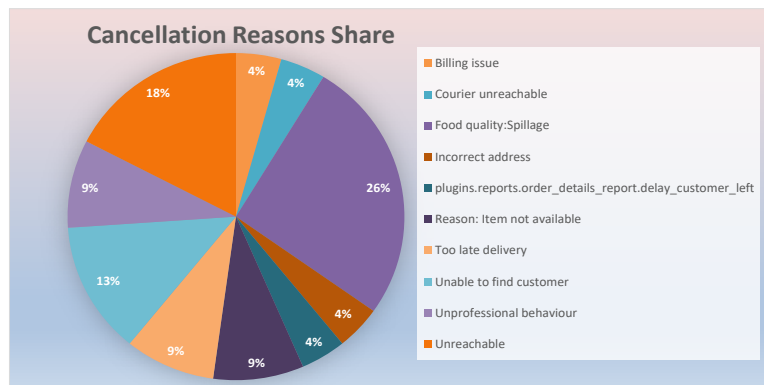


Figure 3.8: Pie Chart Showing Cancellation Reasons Share

The above chart shows that though there were only 23 cancellations total in September and October combined, the cancellation breakdown shows a major internal quality problem and significant delivery/communication weaknesses. Food spillage is the single largest cause at 26%, indicating packaging or kitchen handling issues that directly damage order quality. Communication and courier execution issues also account for a large share: Unreachable customers (18%) and Unable to find customer (13%), plus Too late delivery (9%), Courier unreachable (4%), and Incorrect address (4%), together these courier/coordination failures sum to 48% of cancellations. Other internal problems include Item not available (9%) and Unprofessional behavior (9%), while technical/plugin-related cancellations sit at 4% and billing issues at 4%. The data indicates a two-pronged priority: fix packaging/food-handling to reduce the 26% spillage loss, and improve delivery coordination to cut nearly half of cancellations.

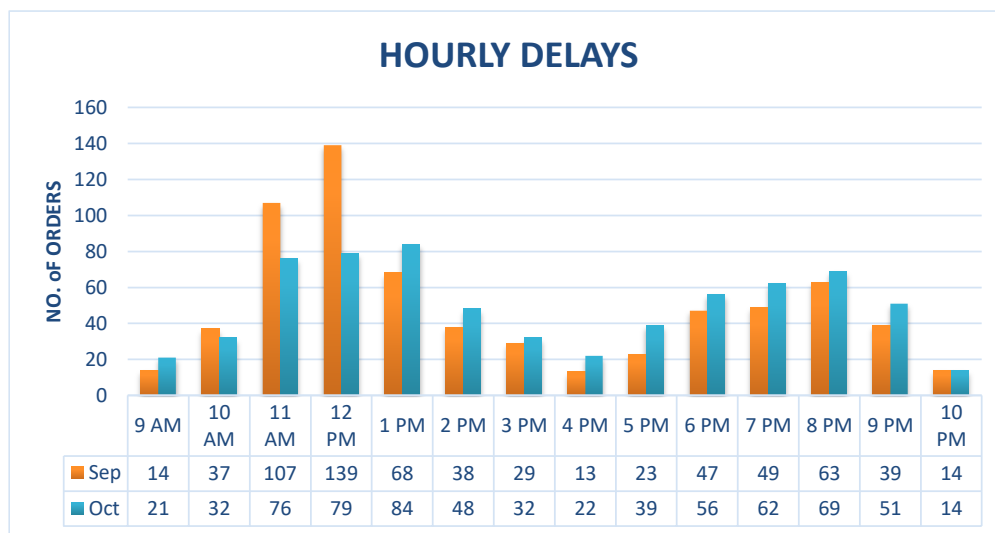


Figure 3.9: Column Chart Showing Hours with Most Preparation Delayed Orders

This chart isolates the specific times when delayed orders were most frequent across September and October. The primary peak for delayed orders occurs during the Lunch Rush (11 AM - 1 PM), with the highest single point being 12 PM in September with 139 delayed orders. A crucial trend is visible month-over-month: the number of delayed orders significantly decreased at 11 AM (from 107 to 76) and 12 PM (from 139 to 79) in October. However, this improvement at midday is offset by generally increased delayed orders during the afternoon and evening hours (1 PM to 9 PM). For example, 1 PM delays rose from 68 to 84, and 9 PM delays rose from 39 to 51. This indicates that the operational efficiency challenge has shifted later in the day rather than being entirely resolved.

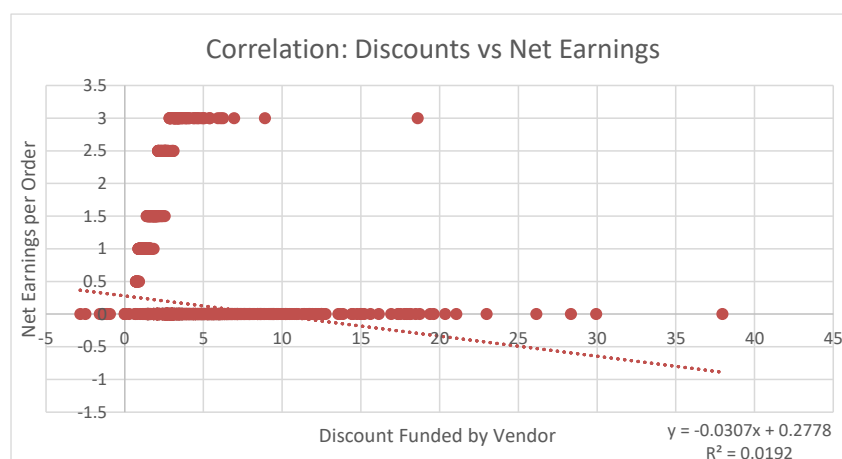


Figure 3.10: Vendor Discount Impact on Net Earnings (Scatter Plot)

The correlation plot comparing Discounts Funded by Vendor with Net Earnings per Order shows an extremely weak and practically non-existent predictive relationship between the two variables. Most data points cluster around zero net earnings, regardless of the discount amount, and the low R-squared value of approximately 0.0192 confirms that vendor-funded discounts are not a significant driver of profitability at the per-order level. Although the trend line indicates a very slight negative correlation, this relationship is neither strong nor consistent enough to be considered reliable. Therefore, changes in discount levels are unlikely to have a meaningful impact on overall net earnings, making vendor-funded discounts statistically ineffective. Should the café stop giving vendor-funded discounts? The café does not necessarily need to stop vendor-funded discounts, because they are not harming profit. But they also do not improve it, so the café can freely choose to keep, reduce, or adjust them without worrying about profit impact. Should discounts be redirected toward customer retention instead of profit impact? Yes. Since discounts do not improve profit, it is more strategic to use them for customer retention instead.

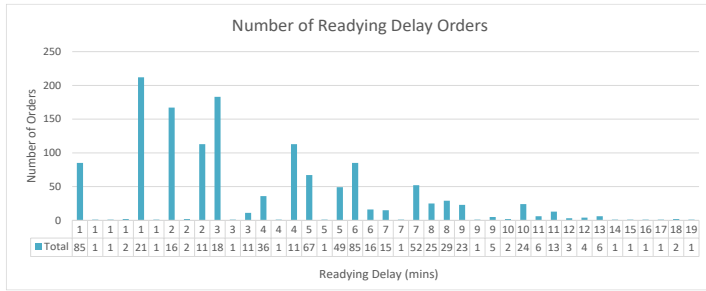


Figure 3.11: Total Ready Delay Orders for Two Months

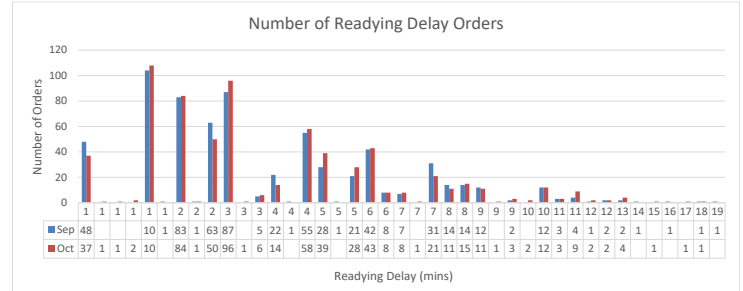


Figure 3.12: Ready Delay Orders Spread

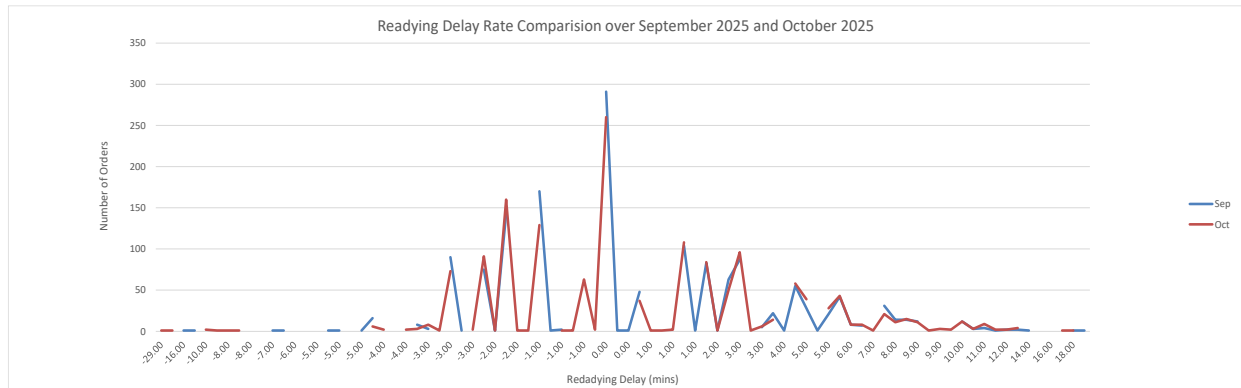


Figure 3.13: Ready Delay Order in Minutes Comparison

This set of charts provides a detailed view of the business's production efficiency and highlights the operational challenges. The key takeaway is that the problem is not extreme delay, but rather the high frequency of minor delays. The data consistently confirms that the greatest concentration of delayed orders occurs within the 1 to 2-minute late window, with the total distribution showing large spikes of orders ready slightly after the expected ready time. This small delay is chronic as this minor delay window accounts for the largest number of delayed transactions, often surpassing 150-200 orders. The comparison between September and October shows that the high frequency of these minor delays has been sustained, if not slightly worsened, indicating the underlying process issue has not been addressed.

Crucially, the distribution of orders is notably skewed towards the positive side of the zero-minute mark, confirming that the majority of orders are being completed after the expected ready time. The peak frequency for orders falls within the 0 to 4 minute delay range. While minor spikes exist in the negative (prepared early) area, these are offset by the consistently high number of orders that are ready just 1 to 3 minutes late, confirming that the operational clock is consistently running behind the promise made to the external platform.

This frequent 1-2 minute lag is a significant factor in frustrating customers and platform couriers, demonstrating a clear gap between current kitchen output speed and platform expectations.

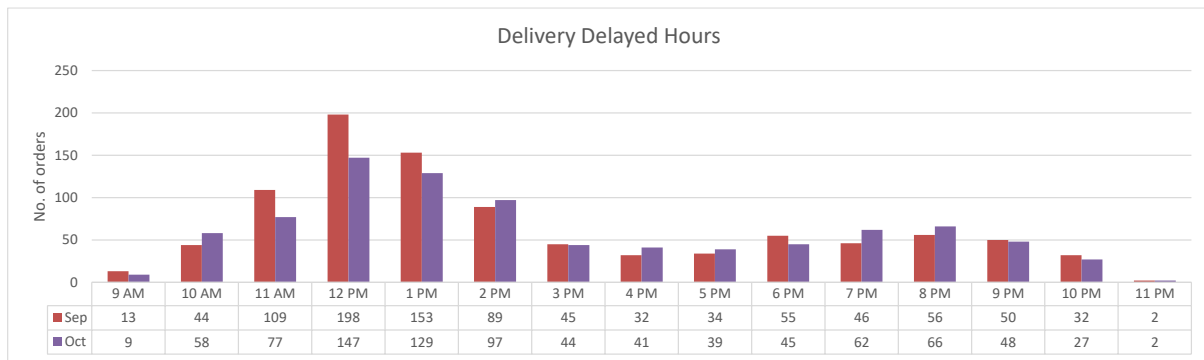


Figure 3.14: Delivery Delayed Hours

This column chart tracks the count of delayed orders, by hour, across September and October, showing that final delivery delay peaks sharply during the Lunch Rush (12 PM), which registered 198 delayed orders in September and 147 in October. A significant operational trend is the decrease in delayed orders during the 11 AM and 12 PM peaks in October, suggesting either improved handling or a direct correlation with the previously identified drop in overall order volume during those hours. Conversely, the frequency of delayed orders generally increased during the evening hours (7 PM and 8 PM), rising from 46 to 62 and 56 to 66 orders, respectively. This confirms that while pressure was relieved at midday, the challenge of achieving timely final delivery has shifted to the later half of the operational day.

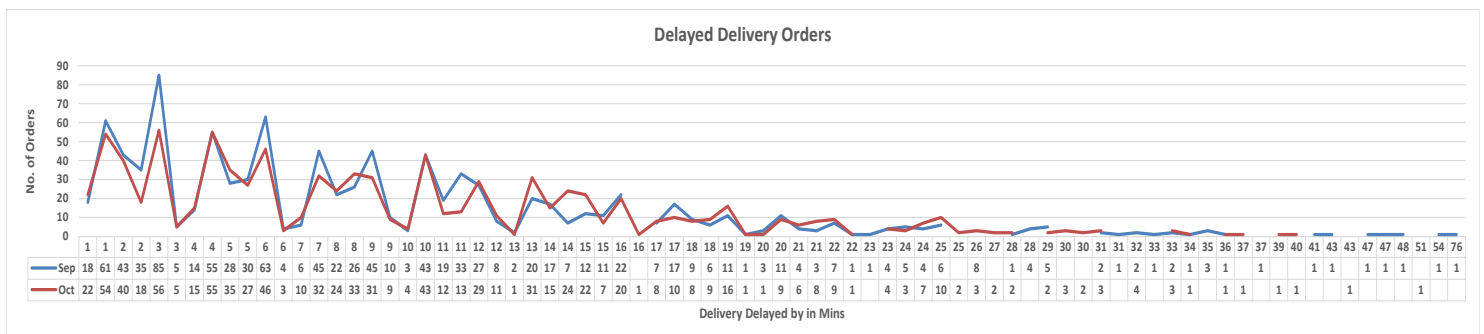


Figure 3.15: Delivery Delayed by in Minutes

This line chart is the most specific breakdown of the long-tail distribution of delivery lateness, measured in minute-by-minute increments.

The data confirms that the overwhelming majority of delivery delays occur in the very short term (1 to 6 minutes), with the highest peak in October occurring at the 2-minute delay mark (54 orders). The frequency rapidly declines after the 6-minute mark, although delays are recorded all the way up to 76 minutes. The comparison between September and October shows a reduction in the number of delayed orders in the 1- 6 minute range, which is the most common customer experience. This suggests a slight improvement in the handling of short delays, which is a positive operational signal. However, the consistent presence of a long tail, with individual orders delayed by 30, 40, or even 50 minutes, indicates that while the common issues are being mitigated, the system still occasionally suffers severe logistical failures out of the café's control as deliveries are completely handled by Talabat.



Figure 3.16: Share of Satisfaction Categories

The pie chart in Figure 3.16 indicates that the customers who receive and review their orders are overwhelmingly satisfied with the experience. A massive 70% of reviewers rated the service as "Amazing". Combining all positive categories ("Good" at 7%, "Very good" at 8%, and "Amazing" at 70%) reveals that 85% of the feedback is highly positive. The remaining 15% is categorized as "OK," which serves as the only significant area for improvement. This overall high satisfaction level is a positive outcome, but it must be viewed cautiously alongside the previously identified operational failures, such as frequent minor delays and food spillage, suggesting that the brand equity built through the product quality is high enough to absorb a significant amount of logistical friction.

The Pie chart in Figure 3.17 reveals that the vast majority of customer dissatisfaction stems from internal production and fulfillment issues, directly impacting the core product quality. The dominant complaint category is Taste/Quality at 41%, closely followed by Wrong/Missing Order complaints at 29%. Together, these two categories account for 70% of all recorded complaints, far exceeding minor issues like Missing Cutlery (13%) or Price Complaints (7%).

Less Quantity complaints are also significant at 10% share. This heavy concentration in quality and accuracy signifies a crucial internal vulnerability: while the brand currently enjoys high overall satisfaction, the high frequency of errors in food quality and fulfillment accuracy poses the most significant long-term threat to the brand's reputation and customer retention.

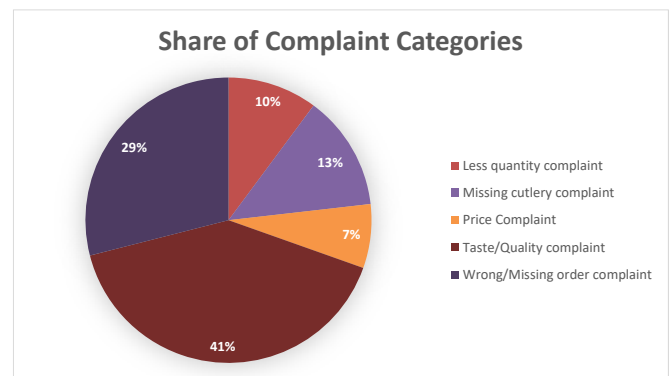


Figure 3.17: Share of Complaint Categories



Figure 3.18: Repeat vs. One-Time Reviewers

The pie chart in Figure 3.18 reveals a significant dynamic in the feedback mechanism: the reviewer base is overwhelmingly dominated by one-time reviewers at 72%. While the 28% segment that has left a review more than once represents a substantial pool of loyal and engaged customers, the high proportion of single reviews suggests that the overall satisfaction scores are heavily influenced by the immediate, transactional experience of a broad and constantly changing audience. This indicates that customer satisfaction, as measured by reviews, is highly dependent on the success of each individual transaction and is less reflective of deep, long-term customer loyalty built through multiple engagements.

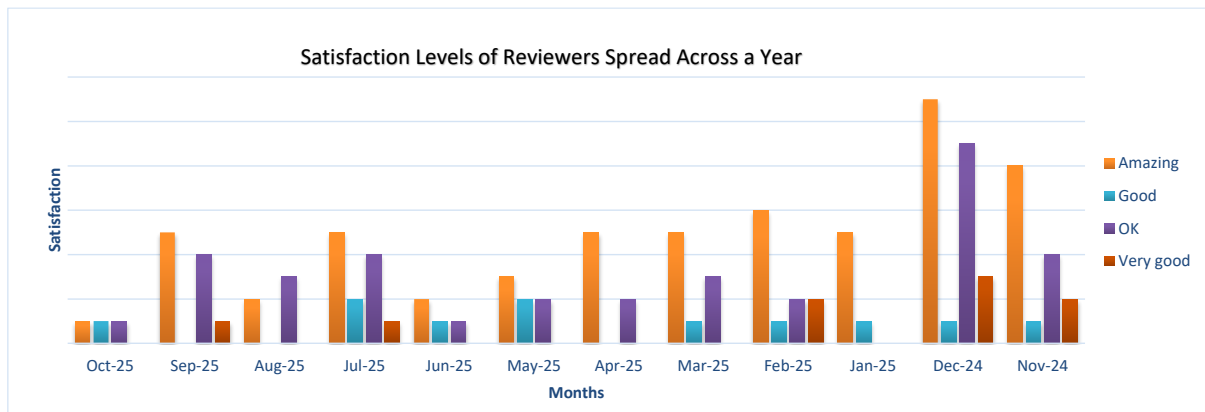


Figure 3.19: Satisfaction Spread across a Year

This time-series column chart provides the long-term context for the customer feedback, revealing a worrying trend in the quality of satisfaction. While the business consistently maintained high positive ratings, there has been a clear and sustained degradation since the high point in late 2024. The data shows a decisive shift away from top-tier ratings ("Amazing" and "Very good") toward the lower-tier categories of "Good" and "OK." This trend accelerated into the recent months of September and October 2025, where the majority of feedback is now concentrated in the "Good" and "OK" bins. This shift confirms that the chronic operational friction, the delays, spillage, and high logistical frustration identified in recent months, is beginning to erode the premium customer experience, moving the business from generating high praise to merely achieving acceptable service levels.

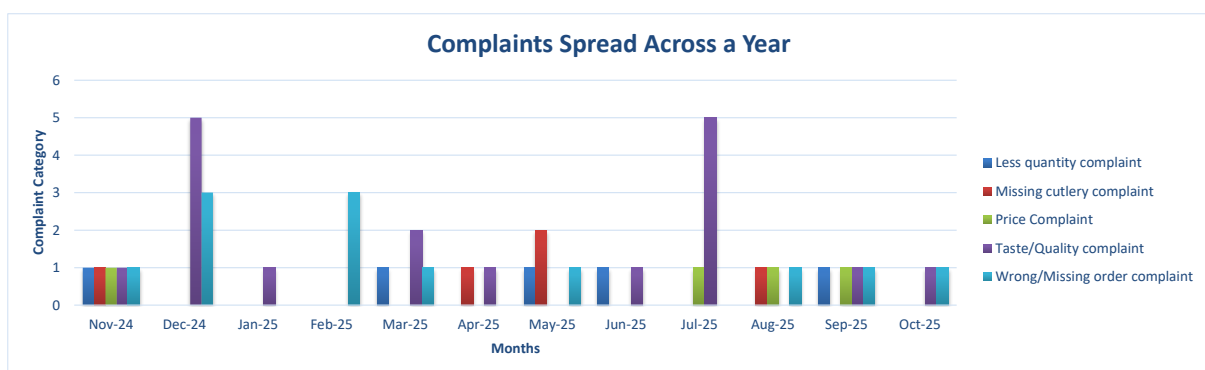


Figure 3.20: Complaint Categories Spread across a year

This time-series column chart provides the historical frequency of different issue types, confirming that while core product errors dominate the complaint share, the absolute number of complaints per month is generally low and episodic. The two dominant complaint categories, Taste/Quality (purple) and Wrong/Missing Order (teal), showed high peaks in specific periods, such as in both Dec-24 and Jul-25. However, outside of these sharp peaks, the total monthly complaint volume remains low and the most recent months (August through October 2025) show low and widely scattered complaints. This suggests that while the severity of the quality control issue is high when it manifests, the frequency of recording these complaints has recently dropped, which is a minor positive trend.

4. Interpretation of Results and Recommendations

4.1 Interpretation of Talabat Expense Analysis

Commission Burden

The financial analysis revealed a critical structural weakness in Sprout's business model, which is an overwhelming dependence on the external delivery platform as discussed in the analysis process, which currently accounts for 99% of all revenue-generating orders.

This dependence is reflected not only in order volume but also in the cost structure, where the platform's commissions and marketing charges consume a substantial portion of Sprout's gross earnings. On average, Sprout retains only 63% of the subtotal value per order, indicating significant margin erosion. When modelling a shift toward direct delivery, the potential improvement becomes stark. Excluding advertising and marketing fees, Sprout could retain up to 90% of the subtotal amount and if such fees are added as well, the retention could increase to 97%. This demonstrates that the platform's commission structure is the single largest drain on the café's profitability. The expense breakdown further strengthens this conclusion.

Commission charges alone reached approximately OMR 4,467 during the analysis period, far exceeding all other expenses combined. Additional spending on Ads Fees and Marketing Fees (totaling OMR 1,235) illustrates that visibility within the app itself requires further investment, undermining profitability even when order volumes are high. These results show that the business is effectively paying twice: once for access to the platform, and again to remain competitive within it.

Subscribed Customers

The Pro vs. Non-Pro customer segmentation reinforces the dependency risk. With 73% of unique customers belonging to the Pro subscription category, demand is being driven by benefits offered by the platform, not by direct loyalty to Sprout. This means Sprout's customer base is effectively borrowed from Talabat, and their purchasing incentives are shaped by external pricing structures outside Sprout's control.

Conclusion

Overall, the financial interpretation clearly indicates that while the external platform is currently drives the engine of sales volume, and also gives visibility to potential customers, it simultaneously imposes significant structural, financial, and strategic constraints, leaving the business vulnerable to fee changes, algorithm shifts, or adverse platform policies and of course, significantly squeezed profit margins.

4.2 Interpretation of Business Performance Analysis

Order Timing and Operational Pressure

The analysis of order distribution across the day identifies two clear operational peaks: Lunch Rush (11 AM-12 PM), the dominant window, generating the highest hourly volumes & Dinner Rush (7 PM-9 PM), a secondary but consistent peak.

A major insight is the sharp drop in lunch-time orders during October. Since these hours contribute the highest proportion of daily demand, even small fluctuations in this window disproportionately impact total monthly revenue. The data confirms that the decline in Sprout's performance is not broad-based but concentrated within its most critical hours.

The quietest periods remain mid-afternoon (3-5 PM) and late evening (9-10 PM), suggesting predictable low-activity windows that could be used strategically for prep, staffing adjustments, or batch production.

Cancellation Rate and Reasons

With a cancellation rate of only 0.74%, the business appears to perform exceptionally well in preventing hard failures.

However, interpretation of cancellation reasons reveals that this metric is somewhat deceptive. Over 65% of cancellations are attributable to operational or logistical shortcomings such as: Customer delay/wait-time frustration, Unreachable or unlocatable customers, Spillage or missing items, Stock unavailability. These are not random failures, they stem from systemic issues either within the kitchen workflow or the delivery network. Thus, while cancellations are low, the underlying causes still represent operational weaknesses that can undermine customer satisfaction and long-term loyalty.

Readying Delay Analysis

A key operational finding is that Sprout suffers primarily from high-frequency minor delays, not major disruptions. Over 1,360 orders were completed late, with the overwhelming majority falling within the 1-2 minute delay window.

The distribution of delays is heavily skewed to the positive side of zero, confirming that the kitchen habitually runs just behind the promised ready time. Although these delays appear minor, the platform's algorithm and customer expectations treat punctuality as a strict metric. This chronic pattern contributes directly to customer wait-time dissatisfaction and aligns with major cancellation categories. The monthly comparison shows no meaningful improvement, indicating that the root operational inefficiency has not been addressed.

Delivery Delay Analysis

Delivery delays follow a similar pattern, the bulk of delays fall within 1-6 minutes, with peaks at 2 minutes. However, the presence of long-tail delays, some extending up to 76 minutes, highlights occasional severe failures in the delivery chain. These outliers, while relatively rare, have outsized impact on customer sentiment and negative reviews. The shift in delayed deliveries from midday (which improved slightly in October) to evening hours suggests that capacity issues or courier shortages, managed by Talabat, may be emerging later in the day.

Conclusion

Overall, the operational interpretation indicates strong performance during off-peak hours but significant inefficiency during high-pressure windows. It can navigate through minimizing readying delay orders by strategic planning but its ability to mitigate delivery delays during high-pressure windows is limited, as these recurring delays are largely dependent on Talabat's drivers and delivery operations, completely outside the café's control.

4.3 Interpretation of Customer Review Analysis

Overall Satisfaction Levels

An encouraging 85% of reviews fall under positive categories (“Amazing”, “Very good”, “Good”), with “Amazing” being the highest at 70%, indicating strong product appeal and general customer approval. However, a worrying trend emerges when examined over a one-year horizon: there is a gradual shift away from top-tier feedback and toward “Good” and “OK” ratings. This decline aligns with the operational issues identified in the dataset, particularly food spillage, delay, and fulfillment inaccuracies.

Complaint Categories

The most significant insight is that 70% of all complaints fall into two categories: Taste/Quality issues (41%) & Wrong or missing items (29%). These are internal process failures, not platform-related issues. Their high frequency indicates inconsistency in kitchen execution, packing processes, or quality checks. Secondary complaints such as missing cutlery (13%) and pricing concerns (7%) are less critical but still signal opportunities for procedural tightening.

Customer Feedback Engagement

Most customers leave a review only once, with very few returning multiple times (maximum observed count: 13 reviews by one customer). This suggests that while Sprout is generating stable positive sentiment, it is not yet building a large, vocal base of repeat reviewers, likely due to the platform controlling the customer relationship.

Conclusion

Customer reviews revealed consistent appreciation for food and service quality, yet highlight concerns about other certain inconveniences, which directly affect satisfaction levels and repeat purchases.

Overall Summary of Interpretation

Collectively, these insights suggest that operational efficiency, demand optimization, and customer experience improvements are essential for sustaining business performance. Across all three analytical dimensions, a consistent theme emerges:

Sprout delivers strong product value but suffers from operational inefficiencies and structural over-reliance on the external delivery platform.

Together, these insights show that Sprout’s performance challenges are not due to poor product quality but **due to process bottlenecks, platform dependence, and a cost structure that limits profitability even at high order volumes.**

Sprout can reduce its reliance on the external platform by adopting its own promotional tactics. Giving exclusive discounts on orders placed through the official website, creating bundle offers available only for direct pickup, and placing flyers or stickers in each bag to advertise its in-house delivery service or website are some strategies it can deploy.

Actionable Recommendations (SMART)

Recommendation 1: Optimize staffing during peak and off-peak hours

- **Specific:** Adjust staff schedules based on hourly order volume patterns.
- **Measurable:** Improve on-time preparation rate to 95%.
- **Achievable:** Use historical hourly demand data.
- **Relevant:** Faster preparation reduces overall delivery time.
- **Time-bound:** Implement within 1 month.

Recommendation 2: Introduce targeted promotions during low-demand hours

- **Specific:** Launch discounts or bundled offers during hours with <30 orders.
- **Measurable:** Increase off-peak orders by 15-20%.
- **Achievable:** Use Talabat in-app promo tools or personally designed pickup offers.
- **Relevant:** Higher order volume offsets platform fees and improves profitability.
- **Time-bound:** Roll out within 2 weeks.

Recommendation 3: Improve rating and review management

- **Specific:** Track repeated complaints & classify repeated complaints into categories.
- **Measurable:** Increase average rating by 0.2 points.
- **Achievable:** Assign one staff member for daily review monitoring.
- **Relevant:** Higher ratings improve ranking and order volume.
- **Time-bound:** Achieve within 2 months.

Recommendation 4: Negotiate commission structure or shift more customers to pickup

- **Specific:** Request revised commission terms or introduce “pickup-only” incentives.
- **Measurable:** Increase pickup orders by 10% or reduce commissions by 2%.
- **Achievable:** Use historical order data to justify negotiation or promote direct pickup.
- **Relevant:** Commission fees significantly impact profitability.
- **Time-bound:** Complete within 1 quarter.

Recommendation 5: Standardize packaging and order-assembling procedures

- **Specific:** Create checklist for packaging and final order verification.
- **Measurable:** Reduce order errors and missing-item complaints by 25%.
- **Achievable:** Incorporate into existing operations.
- **Relevant:** Directly impacts customer satisfaction and review quality.
- **Time-bound:** Apply within 3 weeks.

Summary

Implementing these recommendations will directly improve both operational performance and customer satisfaction. By reducing delays, optimizing staff allocation, and stabilizing order throughput, the business can achieve faster service and higher delivery accuracy. Promotional strategies and commission optimization will strengthen profitability, especially during low-demand periods. Enhancing review management and packaging standards will build customer trust and improve platform visibility, which can increase new and repeat orders. Together, these actions create a sustainable improvement cycle, better operations lead to higher satisfaction, which leads to increased sales and stronger financial performance.